Limits Graphically and Numerically

ANSWER KEY

Section 1: Estimating Limits Numerically & Tables

1 A function g satisfies $g(1.9) = 3.81, g(1.99) = 3.98, g(2.01) = 4.02,$ and $g(2.1) = 4.19$. What is the estimated value of $\lim_{x \to 2} g(x)$?

a. 3.9

b. 4.0 ✓

c. 4.1

d. Does not exist

As x approaches 2 from both the left $(1.9, 1.99)$ and right $(2.1, 2.01)$, the values of $g(x)$ consistently approach 4.0.

2 Based on table values of $f(x) = \frac{\sin(3x)}{x}$ as $x \to 0$ (x in radians), what is the estimated value of $\lim_{x \to 0} f(x)$?

a. 0

b. 1

c. 3 ✓

d. ∞

For $x = 0.01$, $f(0.01) = \frac{\sin(0.03)}{0.01} \approx 2.99955$. The numerical values approach 3.

3 Values of a function $h(x)$ near $x = 3$ are $h(2.99) = 5.01$ and $h(3.01) = -4.99$. What does this suggest about $\lim_{x \to 3} h(x)$?

a. The limit is 0

b. **The limit does not exist because left and right limits disagree** ✓

c. The limit is 5

d. The function is continuous at $x = 3$

Because the left-hand limit approaches $+5$ while the right-hand limit approaches -5, the two-sided limit does not exist.

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Section 2: Graphical Analysis & Removable Discontinuities

4 The graph of $f(x) = \frac{x^2-1}{x-1}$ has a hole at $(1, 2)$. What is the value of $\lim_{x\to 1} f(x)$?

a. 1

b. 2 ✓

c. 0

d. Undefined

Limits describe the behavior near $x = 1$, not at $x = 1$. Since $f(x) = x + 1$ for $x \neq 1$, $\lim_{x\to 1}(x + 1) = 2$.

5 A function f satisfies $\lim_{x\to 4^-} f(x) = 7$, $\lim_{x\to 4^+} f(x) = 7$, and $f(4) = 2$. What is $\lim_{x\to 4} f(x)$?

a. 2

b. 7 ✓

c. 4.5

d. Does not exist

Since both one-sided limits are equal to 7, the two-sided limit is 7, regardless of the defined value $f(4) = 2$.

6 Which of the following functions has a vertical asymptote at $x = 2$ and NOT a removable hole?

a. $f(x) = \frac{x-2}{x^2-4}$

b. $f(x) = \frac{x+1}{(x-2)^2}$ ✓

c. $f(x) = \frac{x^2-4}{x-2}$

d. $f(x) = \frac{2x-4}{x-2}$

In $f(x) = \frac{x+1}{(x-2)^2}$, the factor $(x - 2)$ does not cancel with the numerator, creating a vertical asymptote at $x = 2$.

Section 3: One-Sided Limits & Absolute Value Quotients

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7 Evaluate the one-sided limit: $\lim_{x \to 0^+} \frac{x}{|x|}$.

a. 1 ✓

b. -1

c. 0

d. Does not exist

For $x > 0$, $|x| = x$, so $\frac{x}{|x|} = \frac{x}{x} = 1$. The limit is 1.

8 Evaluate: $\lim_{x \to 5^-} \frac{x-5}{|x-5|}$.

a. 1

b. -1 ✓

c. 0

d. Does not exist

For $x < 5$, $x - 5 < 0$, which means $|x - 5| = -(x - 5)$. Thus $\frac{x-5}{-(x-5)} = -1$.

9 What is $\lim_{x \to 2} \frac{|x-2|}{x-2}$?

a. 1

b. -1

c. 0

d. **Does not exist** ✓

The left-hand limit is -1 and the right-hand limit is 1. Because they are unequal, the two-sided limit does not exist.

Section 4: Infinite Limits & Horizontal Asymptotes

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10 Evaluate: $\lim_{x \to 3^+} \frac{1}{x-3}$.

a. ∞ ✓

b. $-\infty$

c. 0

d. 1

As $x \to 3$ from the right, the denominator is positive and approaches 0, so the quotient grows without bound toward ∞.

11 Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{7x^3 - 4x + 1}{2x^3 + 5x^2 - 3}$.

a. $\frac{7}{2}$ ✓

b. 0

c. ∞

d. $-\frac{1}{3}$

Divide all terms by x^3: $\lim_{x \to \infty} \frac{7 - 4/x^2 + 1/x^3}{2 + 5/x - 3/x^3} = \frac{7}{2}$.

12 What are all horizontal asymptotes of the function $f(x) = \frac{3x}{\sqrt{x^2 + 4}}$?

a. $y = 3$ **and** $y = -3$ ✓

b. $y = 3$ only

c. $y = 0$

d. No horizontal asymptotes

As $x \to \infty$, $\sqrt{x^2} = x$, so $f(x) \to 3$. As $x \to -\infty$, $\sqrt{x^2} = -x$, so $f(x) \to -3$. The asymptotes are $y = \pm 3$.

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